MatrixOne：HTAP数据库中的OLAP设计

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DataFunSummit # 2023

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MatrixOne 简介

矩阵起源是一家数据库创业公司，致力于打造开源超融合异构数据库MatrixOne，MatrixOne是一款面向未来的超融合异构云原生数据库管理系统。通过全新设计和研发的统一分布式数据库引擎，能够同时灵活支持OLTP、OLAP、Streaming等不同工作负载的数据管理和应用，用户可以在公有云、自建数据中心和边缘节点上无缝部署和运行。

Github：https://github.com/matrixorigin/matrixone

目录
CONTENT

01 MatrixOne 整体架构

02 MatrixOne OLAP 引擎

03 TPC-H 测试结果

01

MatrixOne整体架构

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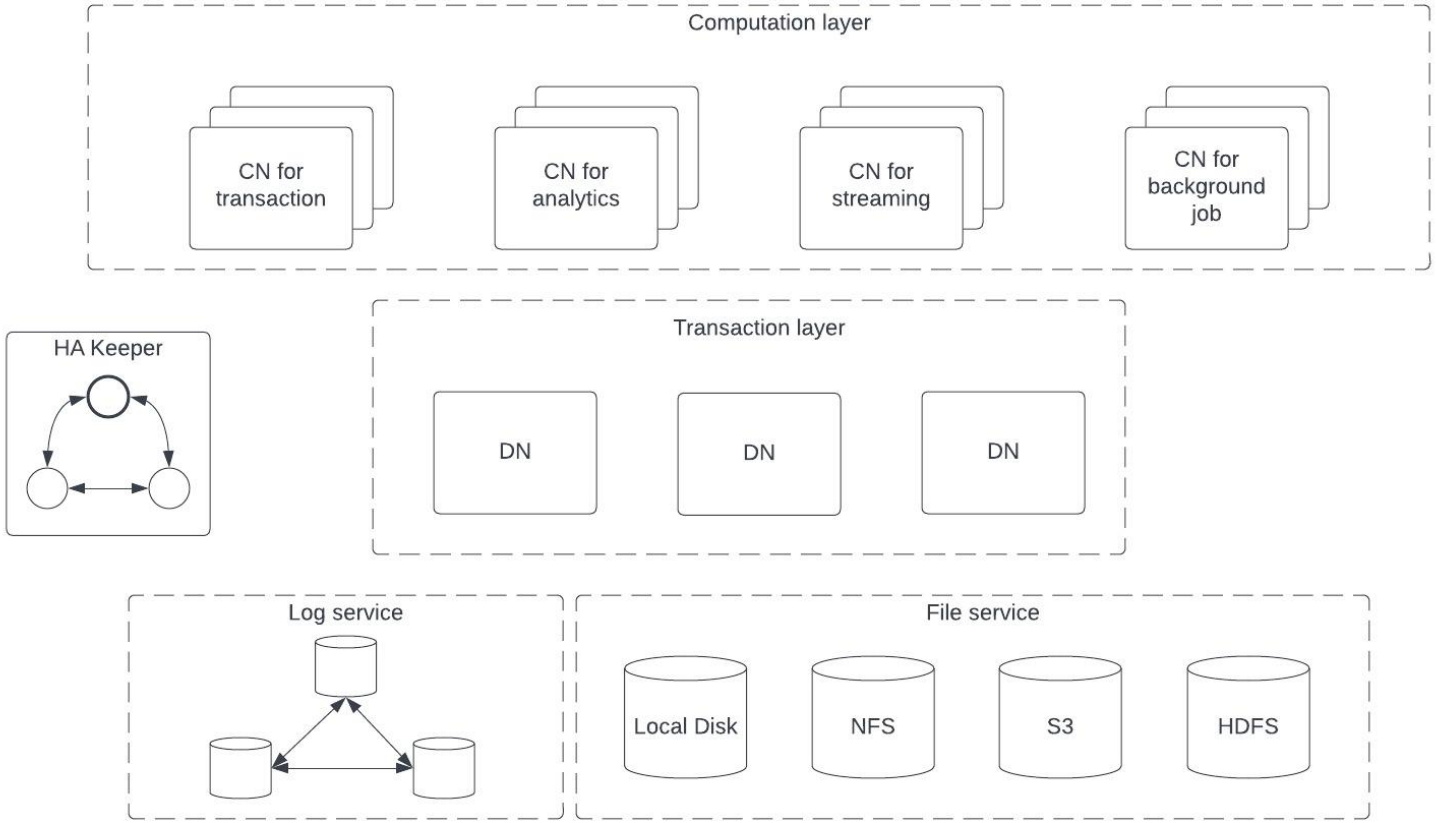
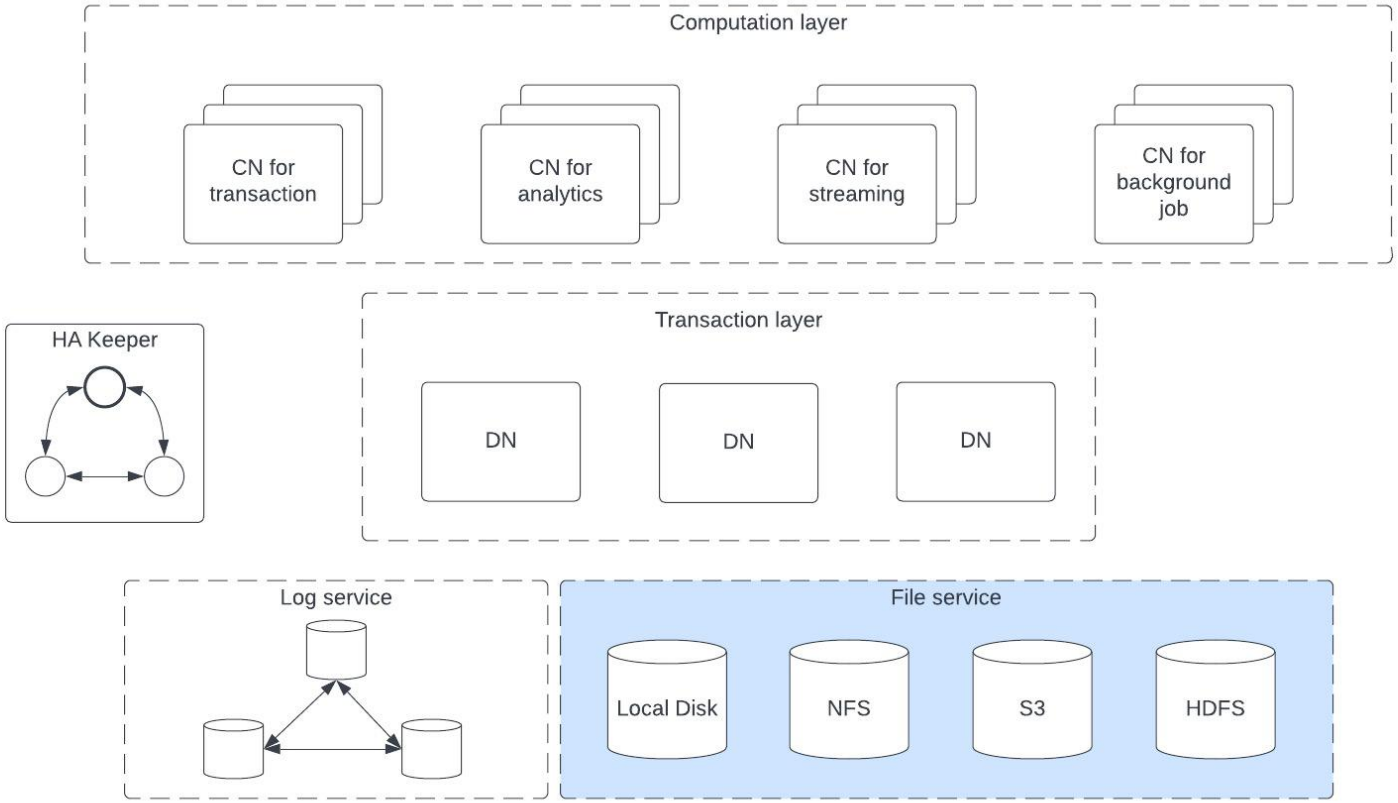
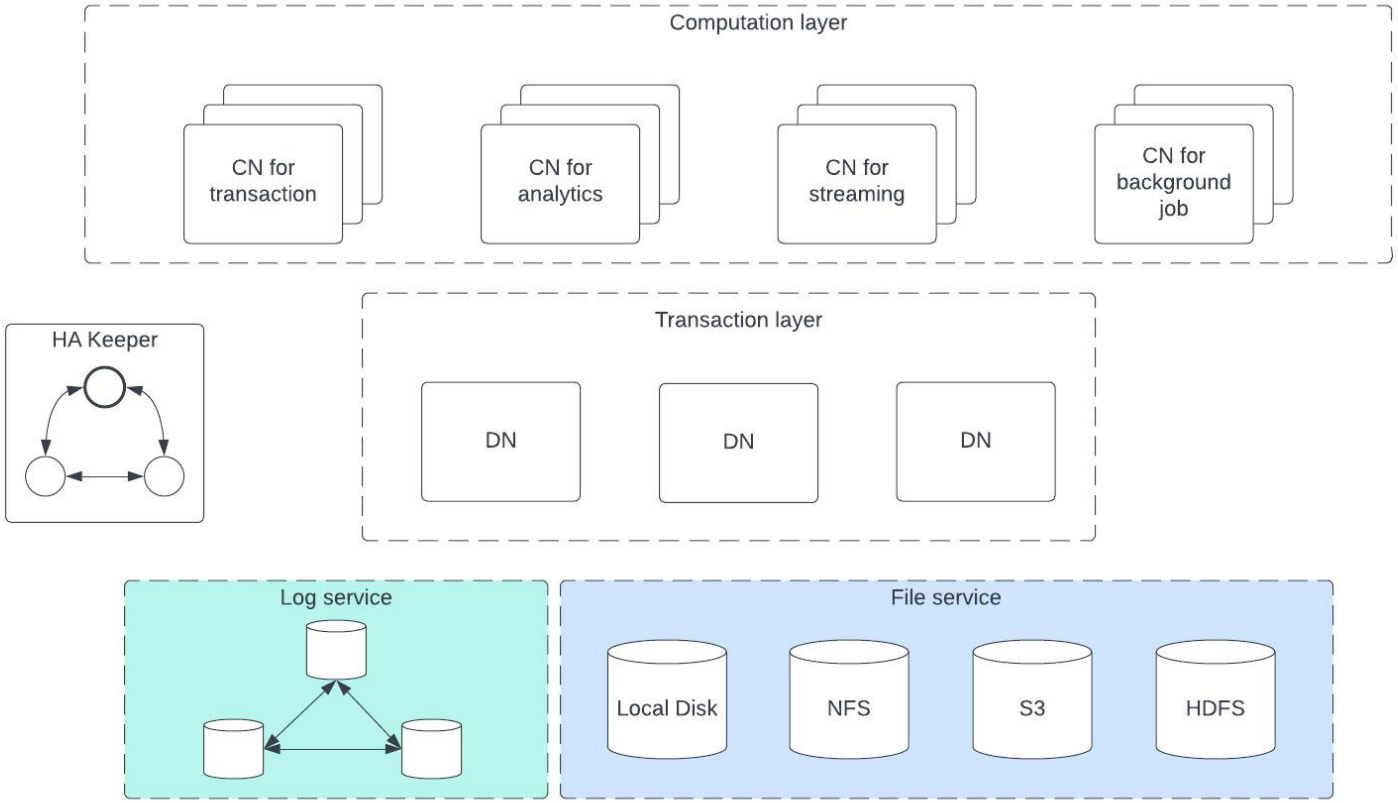
Overview


Computation layer
CN for transaction
CN for analytics
CN for streaming
CN for background job
HA Keeper
Transaction layer
DN
DN
DN
Log service
File service
Local Disk
NFS
S3
HDFS

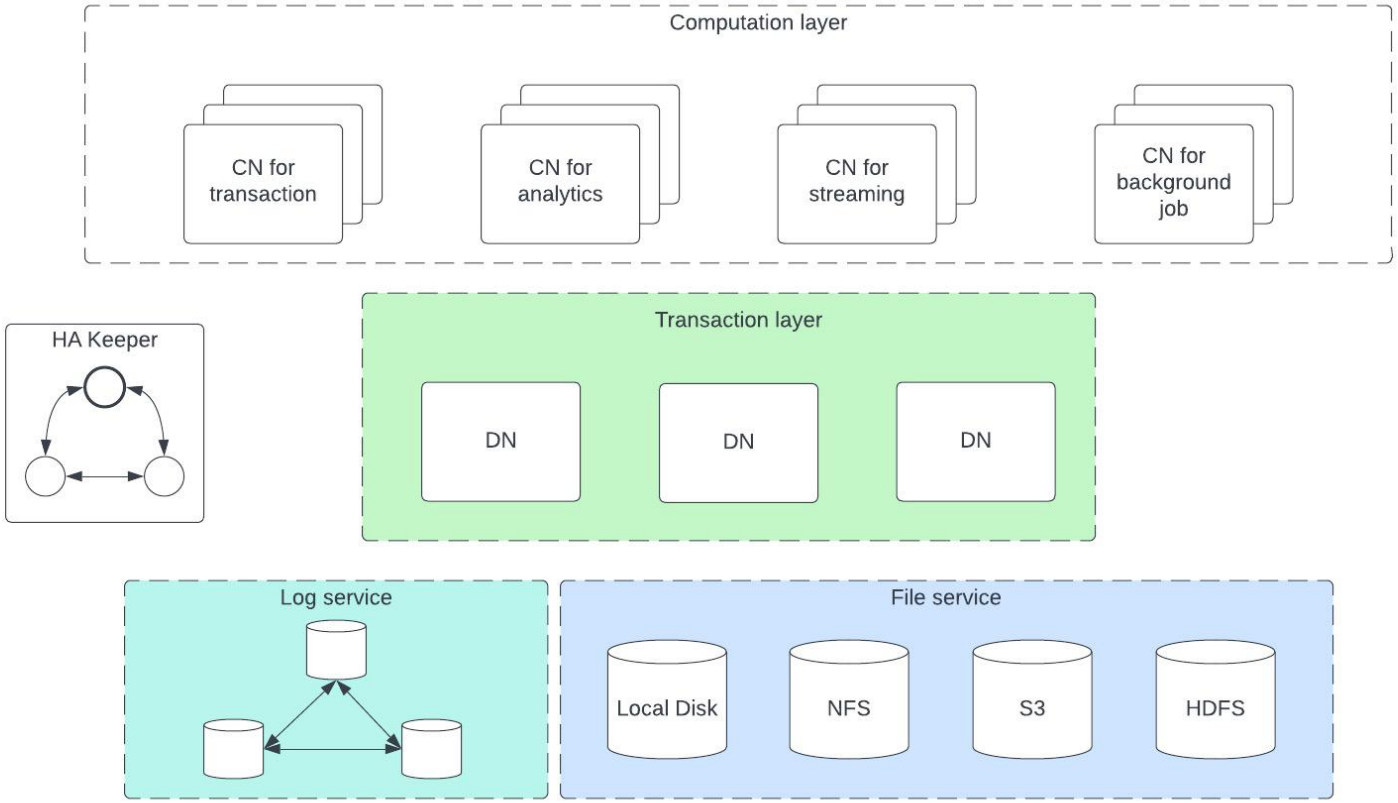
File service



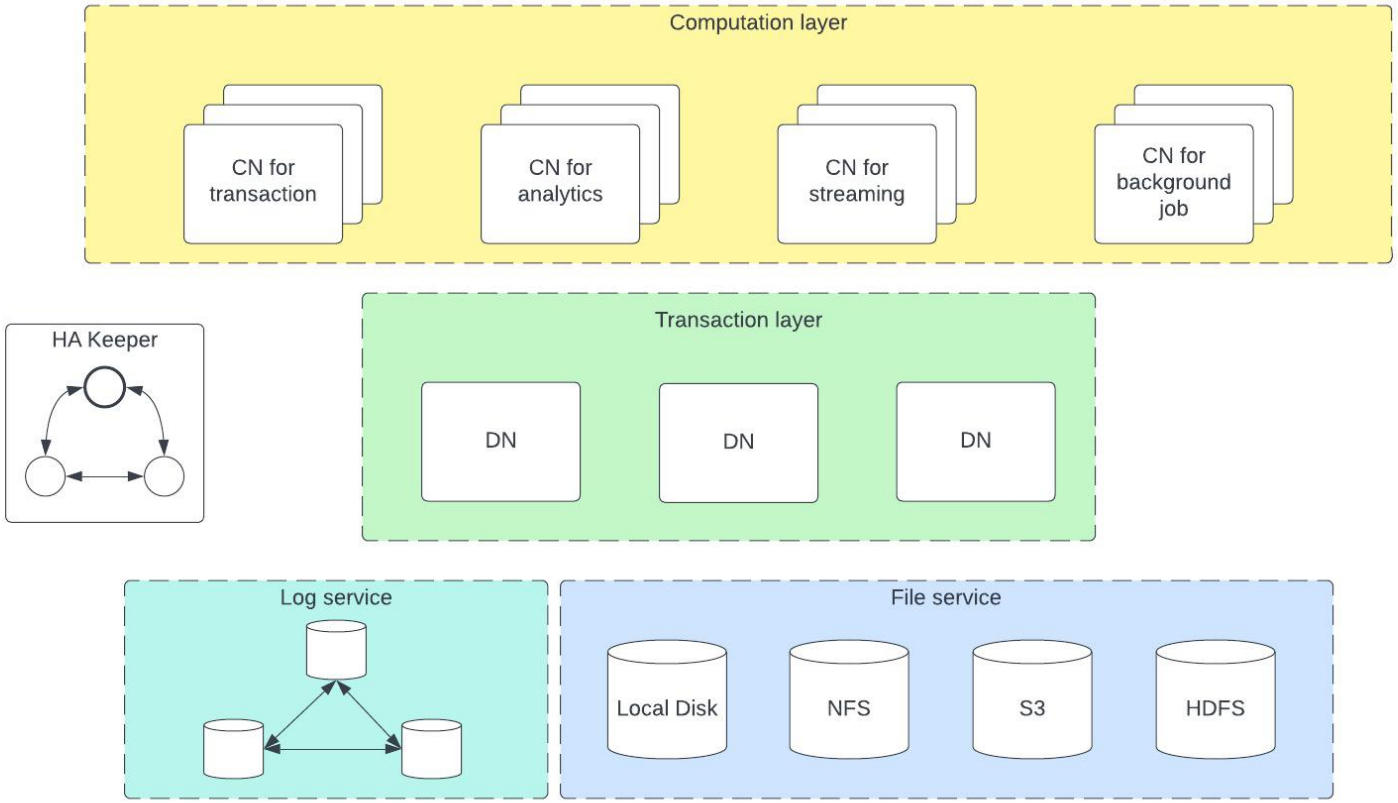
Log service


Computation layer
CN for transaction
CN for analytics
CN for streaming
CN for background job
HA Keeper
Transaction layer
DN
DN
DN
Log service
File service
Local Disk
NFS
S3
HDFS

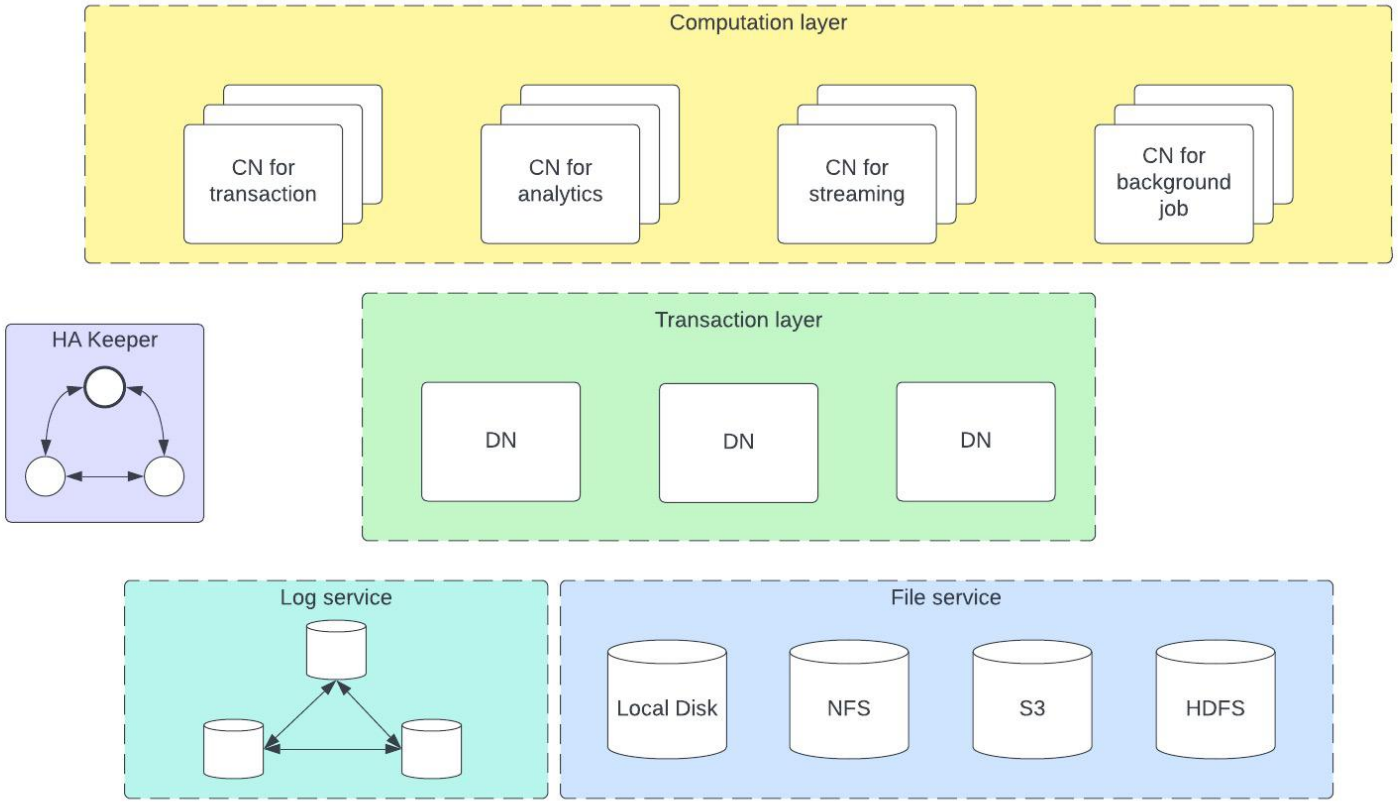
Data node


Computation layer
CN for transaction
CN for analytics
CN for streaming
CN for background job
HA Keeper
Transaction layer
DN
DN
DN
Log service
File service
Local Disk
NFS
S3
HDFS

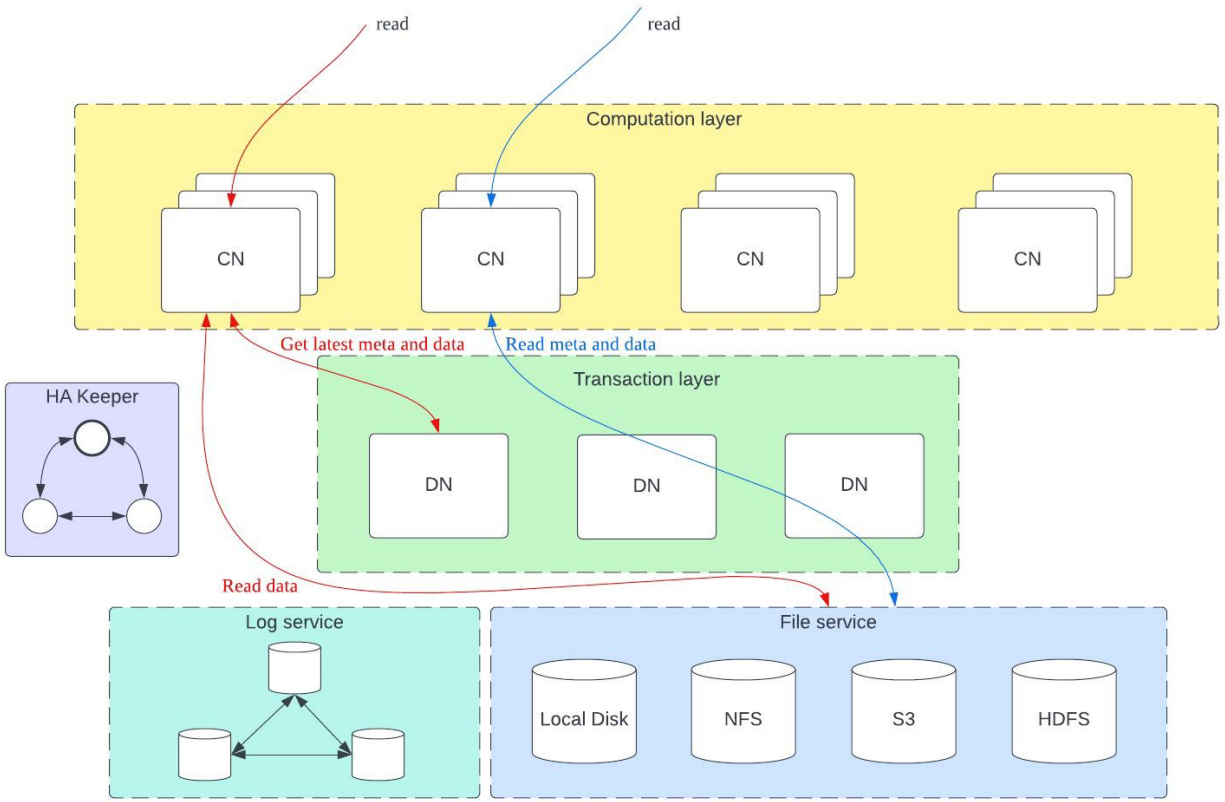
Computation node



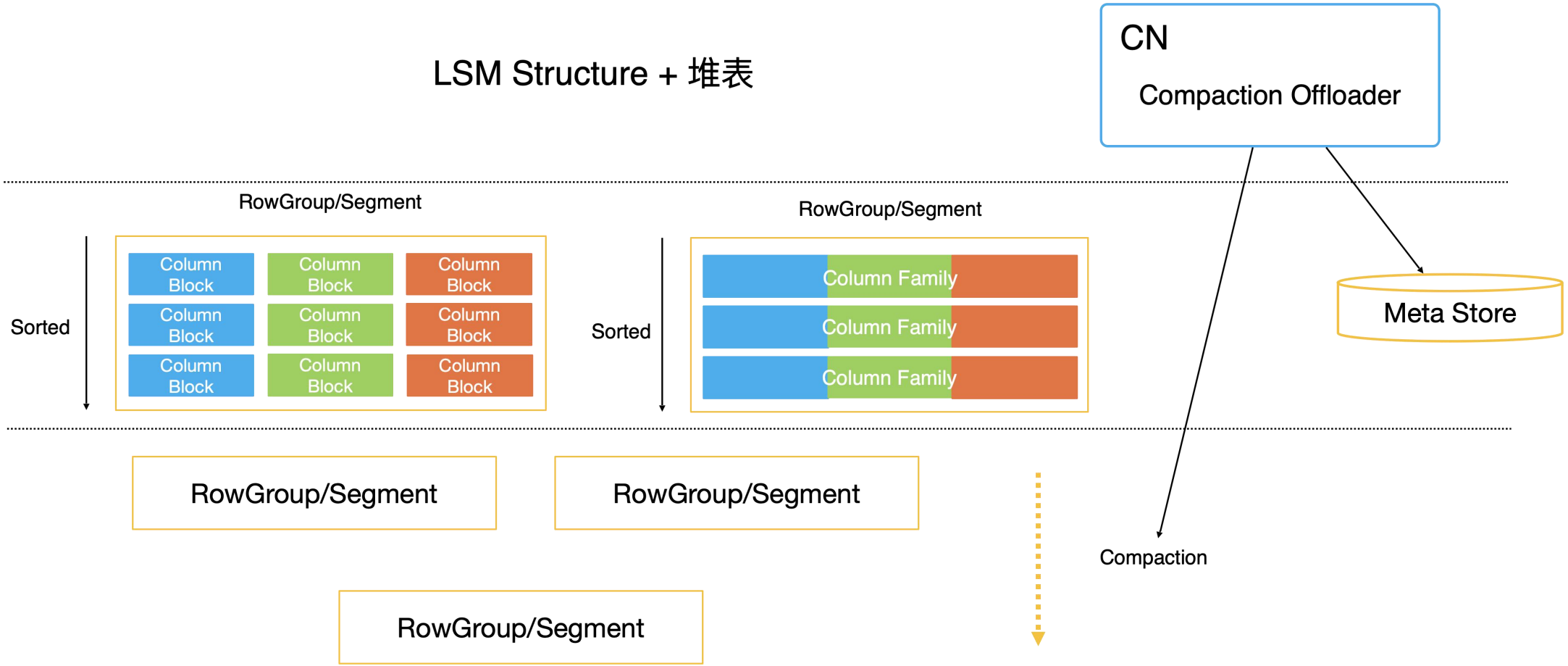
HA keeper


Computation layer
CN for transaction
CN for analytics
CN for streaming
CN for background job
HA Keeper
Transaction layer
DN
DN
DN
Log service
File service
Local Disk
NFS
S3
HDFS

Read data


read
read
Computation layer
CN
CN
CN
CN
Get latest meta and data
Read meta and data
Transaction layer
HA Keeper
DN
DN
DN
Read data
Log service
File service
Local Disk
NFS
S3
HDFS

TAE（Transactional Analytical Engine）



02

MO的OLAP计算引擎

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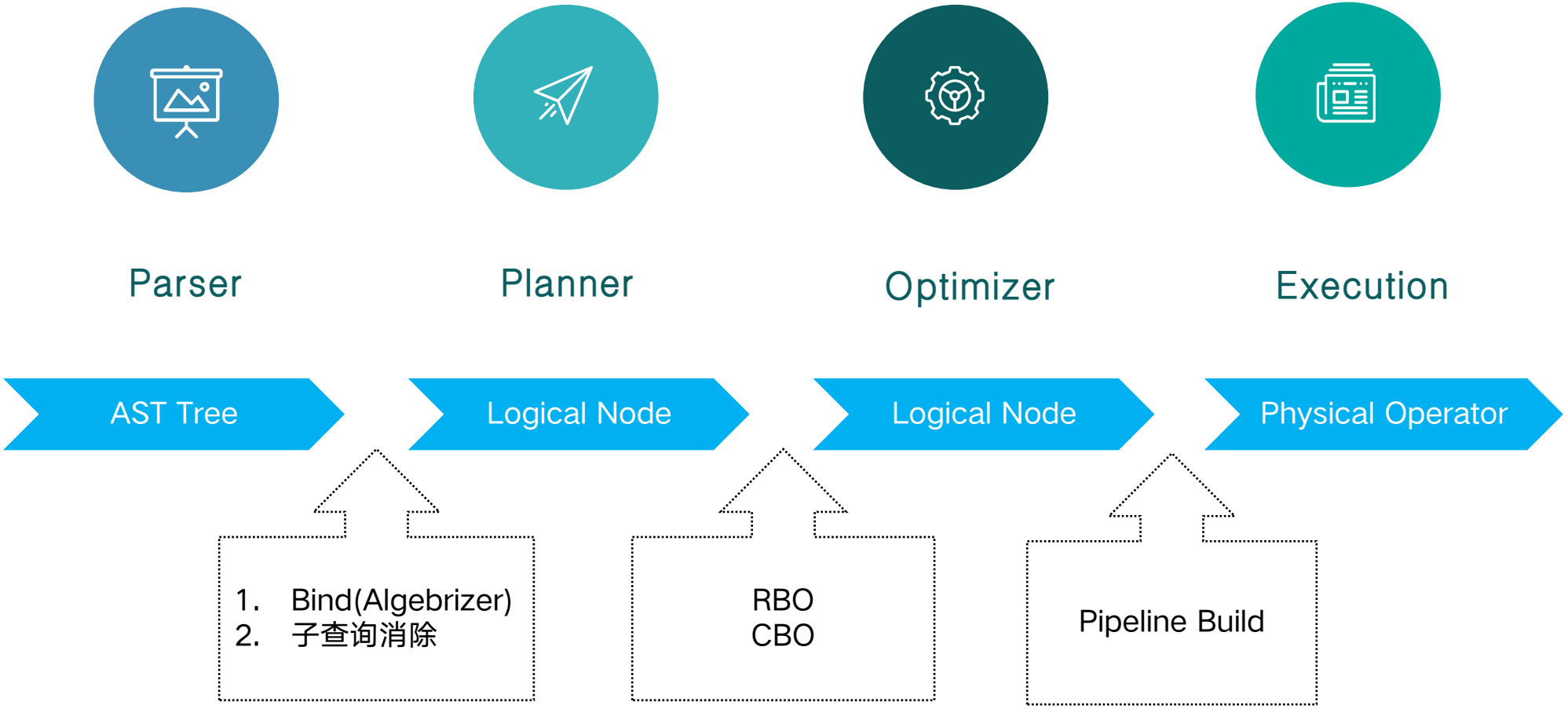
OLAP需要具备的SQL能力

- 多表join
- 子查询
- 窗口函数
- CTE/Recursive CTE
- UDF
- ……

计算引擎构成


Parser
Planner
Optimizer
Execution
AST Tree
Logical Node
Logical Node
Physical Operator
1. Bind(Algebrizer)
2. 子查询消除
RBO
CBO
Pipeline Build

子查询展开

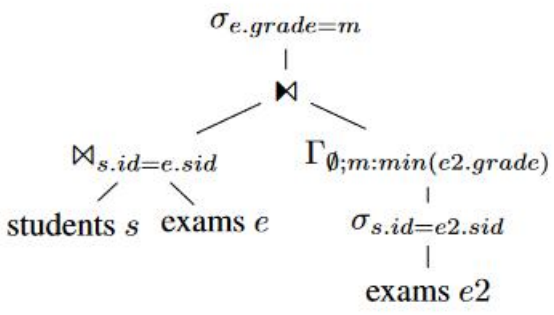
- TPC-H Q2，4，11，16，17，18，20，21，22

子查询展开

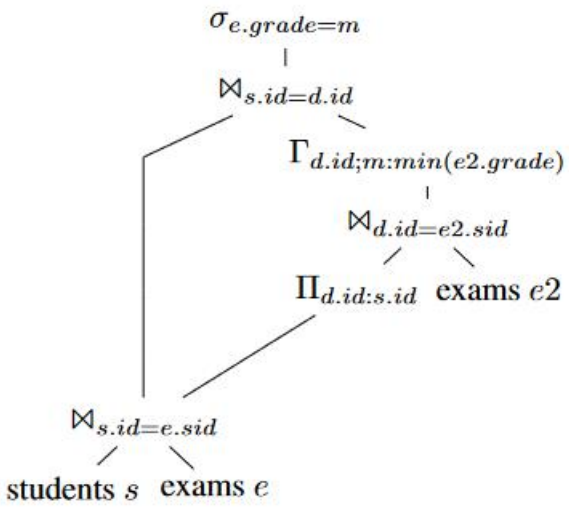
-
```
select s.name,e.course
from students s,exams e
where s.id=e.sid and
e.grade=(select min(e2.grade)
from exams e2
where s.id=e2.sid)
```

子查询展开（HyPer）

展开前



展开后



子查询展开（MatrixOne）

展开前

- ```
  select s.name,e.course
  from students s,exams
  e
  where s.id=e.sid and
  e.grade=(select
  min(e2.grade)
  from exams e2
  where s.id=e2.sid)
  ```

展开后

- ```
  select s.name,e.course
  from students s,exams
  e,(select
  sid,min(e2.grade) as
  min_grade from exams
  e2 group by e2.sid) e3
  where s.id=e.sid and
  e.grade=e3.min_grade
  and s.id=e3.sid
  ```

Optimizer

- 减少IO
  - 列裁剪
  - 谓词下推
  - 谓词推断：q7 q19
  - Runtime filter：q18

Optimizer

- 减少计算
  - Join order：q2 q5 q7 q8 q9 q21
  - 聚合下推/上拉：q13 q17
    - Groupjoin

谓词推断(TPC—H Q19)

```sql
select
    sum(l_extendedprice* (1 - l_discount)) as revenue
from
    lineitem,
    part
where
    (
        p_partkey = l_partkey
        and p_brand = 'Brand#23'
        and p_container in ('SM CASE', 'SM BOX', 'SM PACK', 'SM PKG')
        and l_quantity ≥ 5 and l_quantity ≤ 5 + 10
        and p_size between 1 and 5
        and l_shipmode in ('AIR', 'AIR REG')
        and l_shipinstruct = 'DELIVER IN PERSON'
    )
    or
    (
        p_partkey = l_partkey
        and p_brand = 'Brand#15'
        and p_container in ('MED BAG', 'MED BOX', 'MED PKG', 'MED PACK')
        and l_quantity ≥ 14 and l_quantity ≤ 14 + 10
        and p_size between 1 and 10
        and l_shipmode in ('AIR', 'AIR REG')
        and l_shipinstruct = 'DELIVER IN PERSON'
    )
    or
    (
        p_partkey = l_partkey
        and p_brand = 'Brand#44'
        and p_container in ('LG CASE', 'LG BOX', 'LG PACK', 'LG PKG')
        and l_quantity ≥ 28 and l_quantity ≤ 28 + 10
        and p_size between 1 and 15
        and l_shipmode in ('AIR', 'AIR REG')
        and l_shipinstruct = 'DELIVER IN PERSON'
    );
```

```sql
select
    sum(l_extendedprice* (1 - l_discount)) as revenue
from
    lineitem,
    part
where
    p_partkey = l_partkey
    and p_size ≥ 1
    and l_shipmode in ('AIR', 'AIR REG')
    and l_shipinstruct = 'DELIVER IN PERSON'
    and (
        (
            p_brand = 'Brand#23'
            and p_container in ('SM CASE', 'SM BOX', 'SM PACK', 'SM PKG')
            and l_quantity ≥ 5 and l_quantity ≤ 5 + 10
            and p_size ≤ 5
        )
        or
        (
            p_brand = 'Brand#15'
            and p_container in ('MED BAG', 'MED BOX', 'MED PKG', 'MED PACK')
            and l_quantity ≥ 14 and l_quantity ≤ 14 + 10
            and p_size ≤ 10
        )
        or
        (
            p_brand = 'Brand#44'
            and p_container in ('LG CASE', 'LG BOX', 'LG PACK', 'LG PKG')
            and l_quantity ≥ 28 and l_quantity ≤ 28 + 10
            and p_size ≤ 15
        )
    );
```

Join order(改进的贪心法)

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- 确定事实表与维度表

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- 确定事实表与维度表
- 各事实表与其维度表join成子树

Join order(改进的贪心法)

- 确定事实表与维度表
- 各事实表与其维度表join成子树
  - 先与过滤性好的维度表join

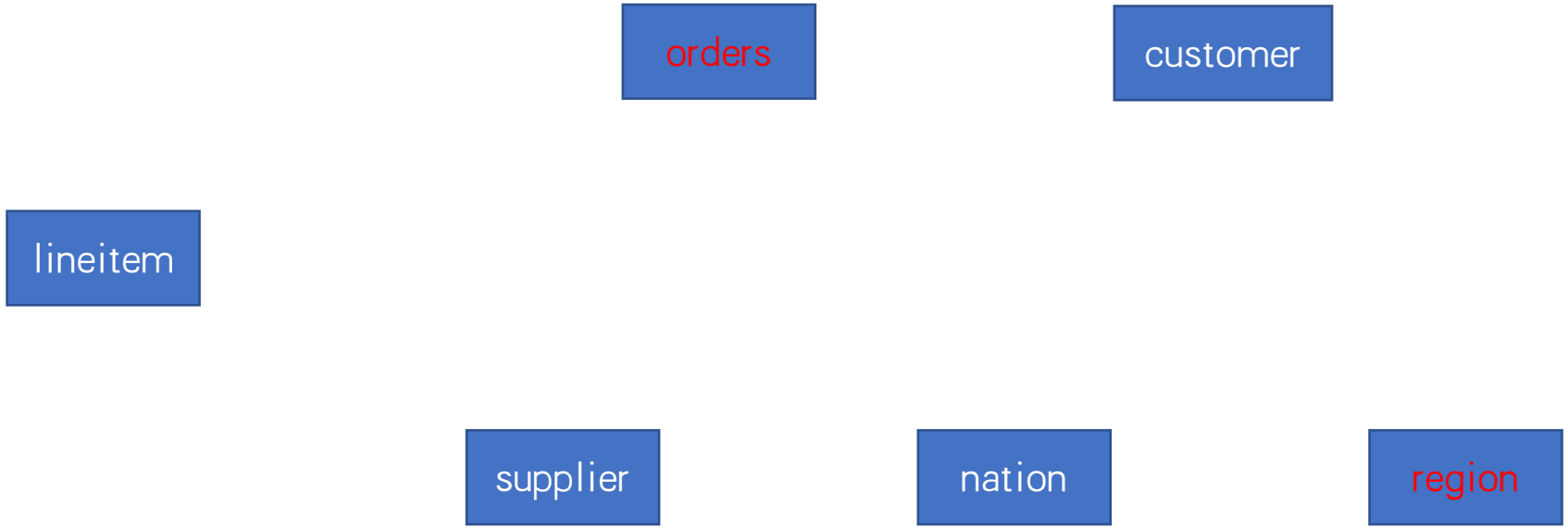
Join order(改进的贪心法)

- 确定事实表与维度表
- 各事实表与其维度表join成子树
  - 先与过滤性好的维度表join
- 子树之间用经典Join order算法

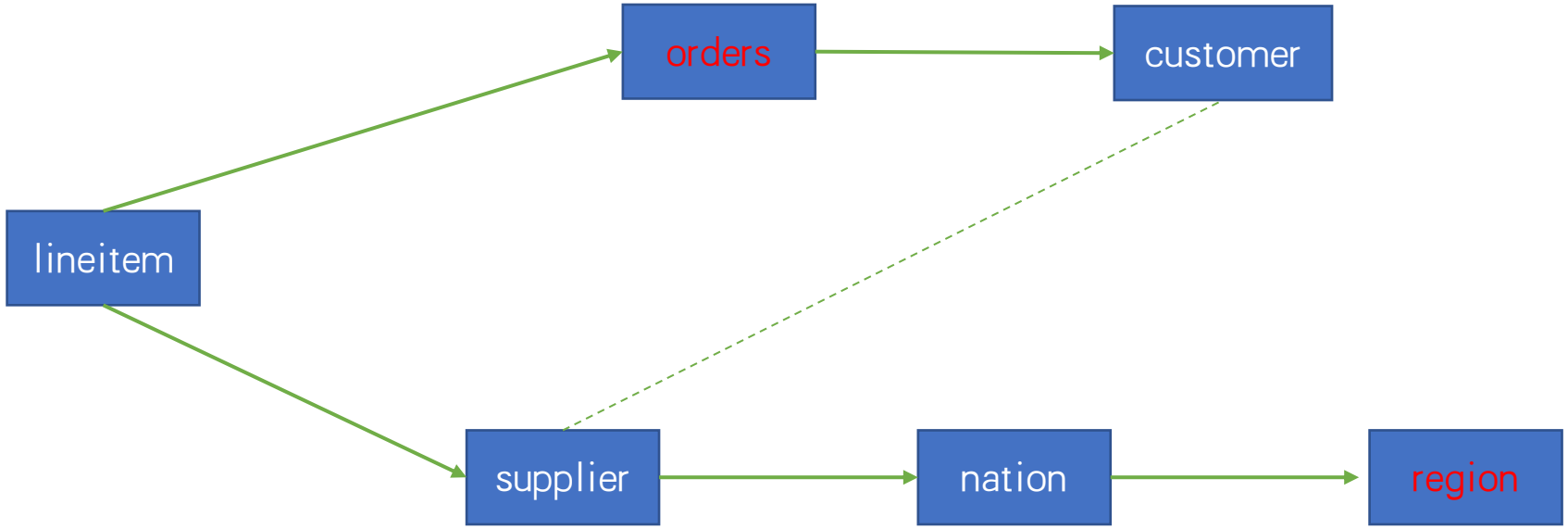
Join order(改进的贪心法)

```sql
select
    n_name,
    sum(l_extendedprice * (1 - l_discount)) as revenue
from
    customer, orders, lineitem, supplier, nation, region
where
    c_custkey = o_custkey and l_orderkey = o_orderkey
    and l_suppkey = s_suppkey and c_nationkey = s_nationkey
    and s_nationkey = n_nationkey and n_regionkey = r_regionkey
    and r_name = 'AMERICA' and o_orderdate ≥ date '1994-01-01'
    and o_orderdate < date '1994-01-01' + interval '1' year
group by
    n_name
order by
    revenue desc
;
```

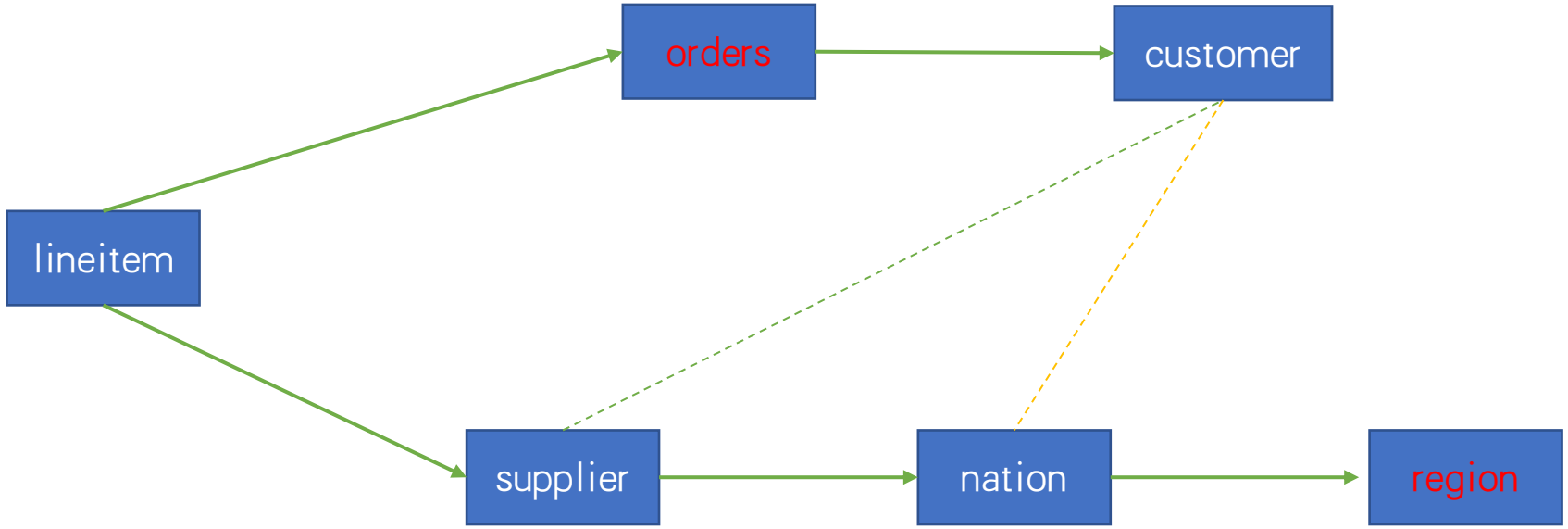
TPC-H Q5



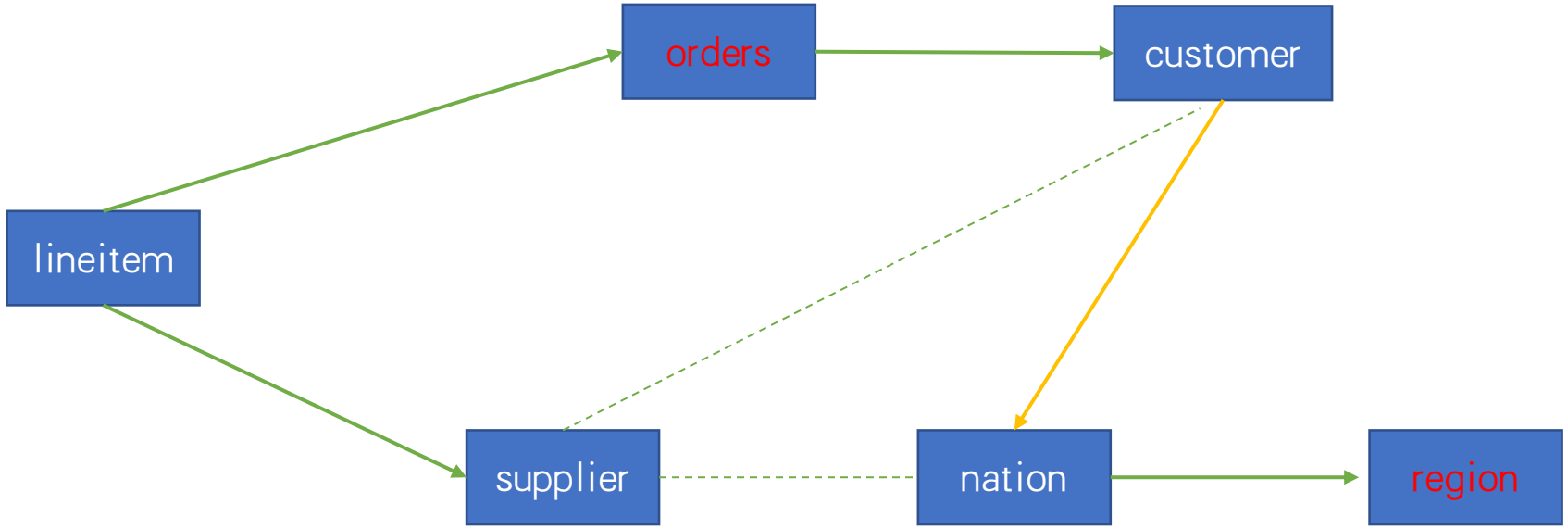
TPC−H Q5



谓词推断 + Join order



谓词推断 + Join order



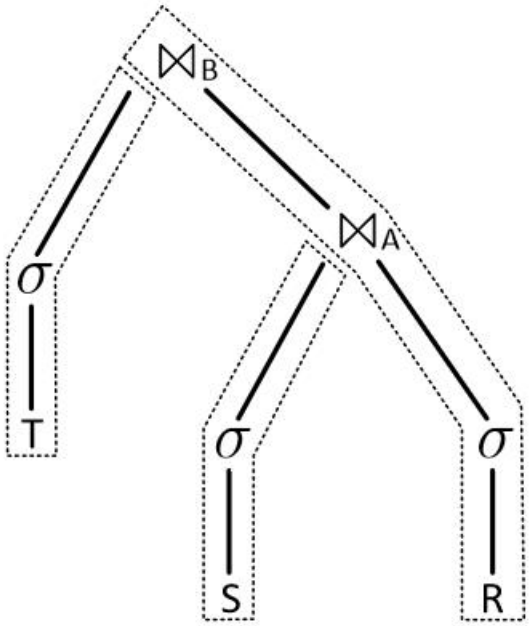
Execution： Pipeline

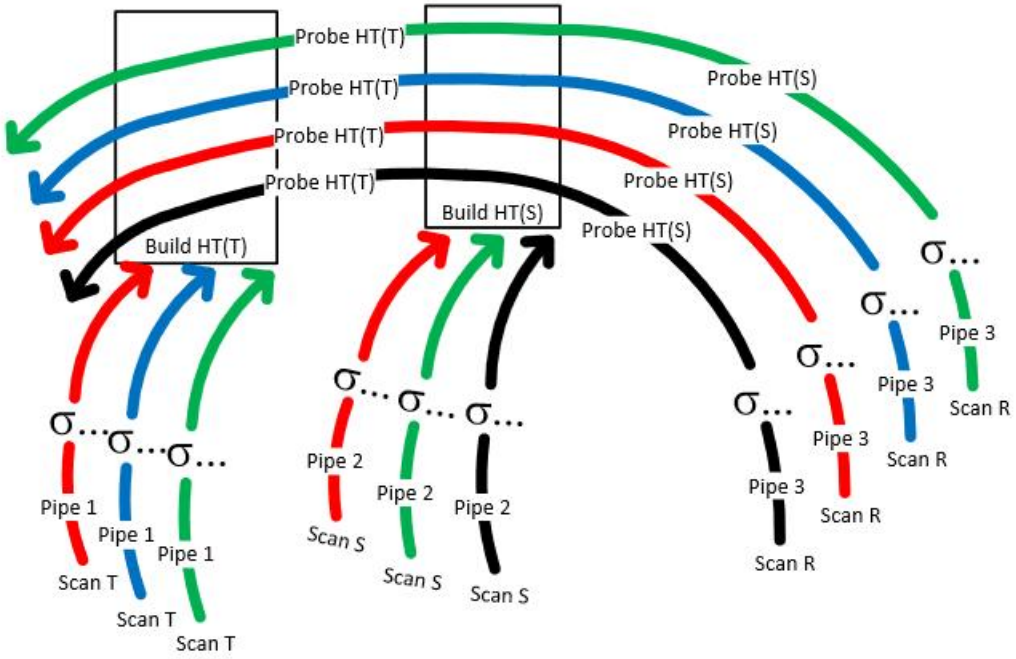
Push模型基于不同的Operator组成流水线Pipeline
自底向上Push调度，以数据为中心而非Operator为中心
Logical Plan 后序遍历即可得到基础Pipeline结构

Morsel-Driven Parallelism
以Task Queue为核心调度Pipeline，每个数据块的单位称为Morsel
一个任务执行结束，通知调度器将后续任务加入队列
CPU空闲时，可以从其他CPU 偷取任务执行(Work Stealing)

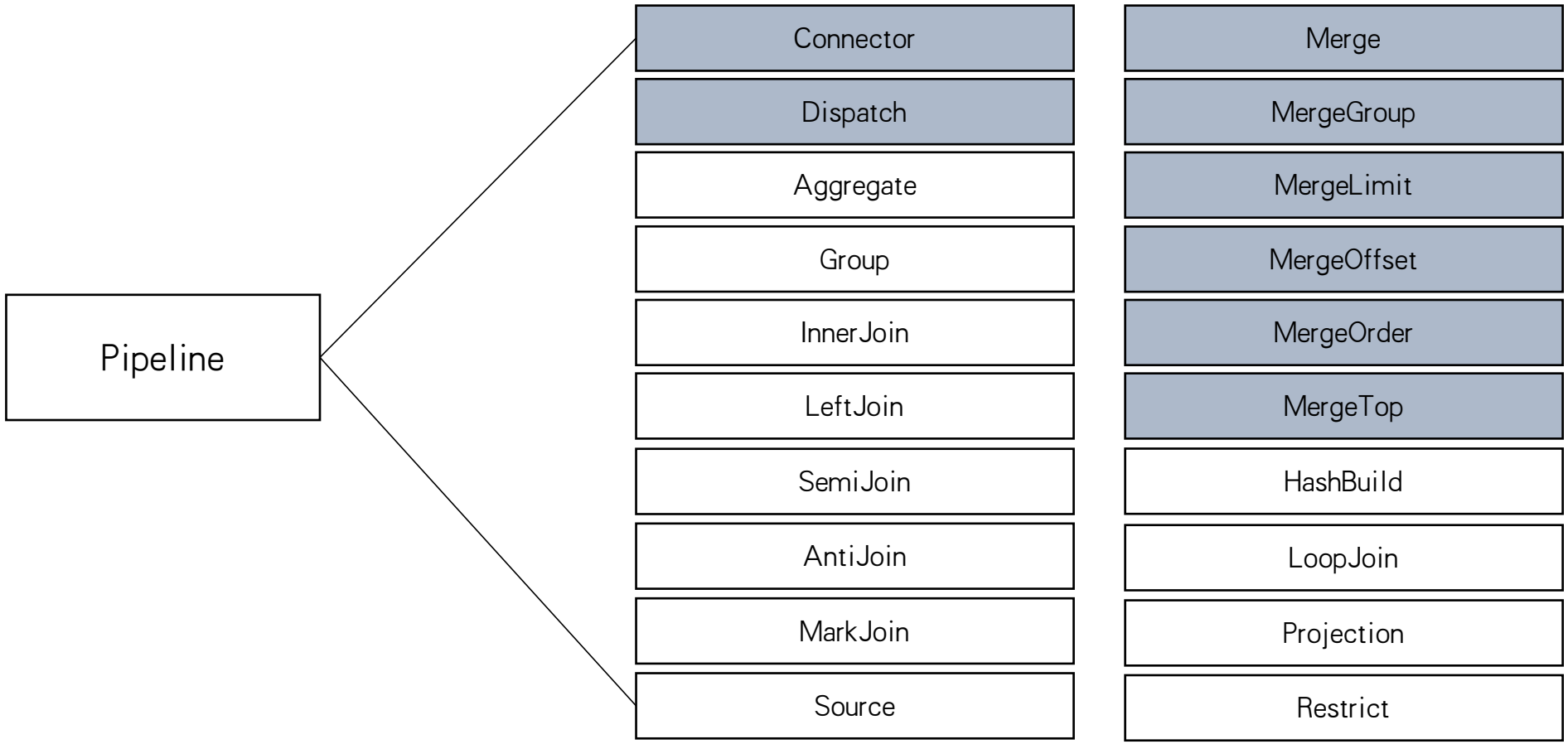
Execution： Pipeline

```
SELECT *
FROM S
JOIN R USING A
JOIN T USING B;
```





Execution：physical operators



03

TPC-H性能测试结果

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TPC-H 100G	SF xs cold(0316)	SF xs hot(0316)	2023/6/8 eks cold 29f1e83	2023/6/8 eks hot 29f1e83
Q1	12	10	67.814	10.75
Q2	4.9	3	18.298	1.75
Q3	5.4	4.5	36.462	10.73
Q4	5.1	5	17.594	10.82
Q5	6.4	6.1	13.817	7.21
Q6	0.49	0.54	1.766	1.21
Q7	4.4	4.1	10.858	10.36
Q8	6.2	5	11.428	5.86
Q9	12	12	40.385	37.1
Q10	10	9.4	20.382	18.46
Q11	2.5	1.9	3.724	1.3
Q12	3.1	2.8	12.29	5.39
Q13	17	15	20.647	20.83
Q14	0.92	0.86	7.04	7.89
Q15	2.7	1.6	3.936	4.34
Q16	3.5	3.2	6.504	5.29
Q17	3.9	3.6	3.473	3.41
Q18	28	25	16.413	12.77
Q19	5.7	3.5	8.002	5.3
Q20	3.2	2.9	2.314	2.55
Q21	12	12	34.46	31.92
Q22	2.3	2.1	3.487	3.8
233.738	151.71	134.10	361.09	219.06

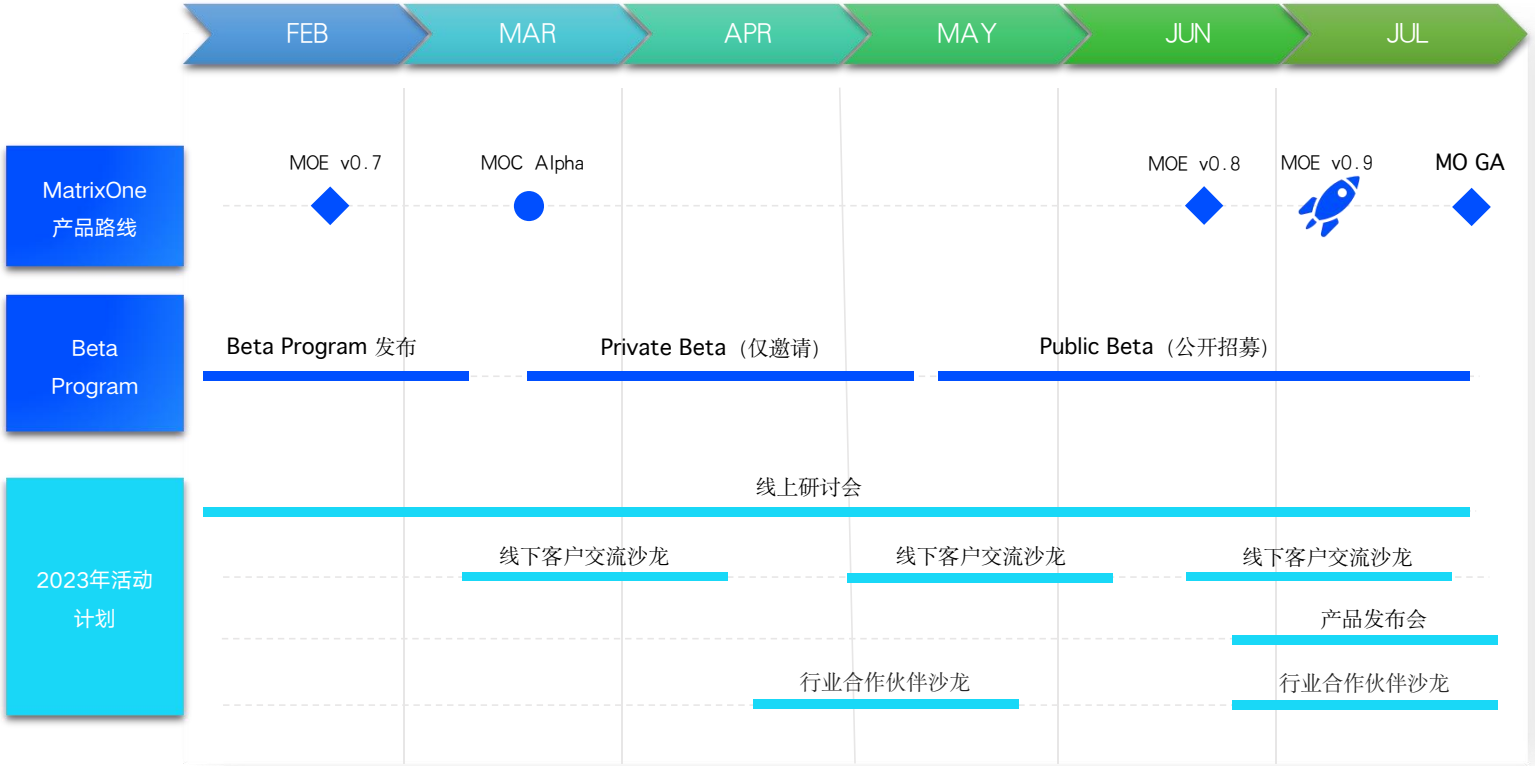
感谢观看

矩阵起源 Matrix origin DataFun.

欢迎加入 MatrixOne Beta Program 用户体验计划

MatrixOne Beta Program 是矩阵起源全新推出的，与客户、用户一起持续提升 MatrixOne 产品和性能体验优化的计划。

- 新功能内测权益
- 产品设计参与权益
- 新功能本地环境优先测试权益
- 开发过程的直接发言权益
- 专家端到端专业支持权益



加入 MatrixOne Beta Program

- Step1：扫描下方小程序码提交注册
- Step2：MO架构师将会通过邮件的方式进行初步联系和沟通
- Step3：加入 Beta Program 社区，开始您和 MatrixOne 的旅程

加入 MatrixOne 社区群，获取 PPT